EntSeg: Entropy-Guided Pseudolabel Denoising and Masked Image Consistency for Cross-Domain Remote Sensing Segmentation

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***Abstract*—Unsupervised domain adaptation (UDA) for semantic segmentation of remote sensing images (RSIs) aims to adapt a model trained on labeled source-domain data to a target domain without requiring target-domain annotations. Existing self-training (ST) approaches adapt the model by generating pseudolabels for target-domain samples. However, due to the inherent domain gap, these pseudolabels often contain substantial noise, which in turn limits the effectiveness of such methods. We observe that noisy pseudolabels typically exhibit higher entropy. Motivated by this observation, we propose an entropy-guided segmentation (EntSeg) framework, a novel entropy-guided framework for cross-domain RSIs semantic segmentation that comprises two key models. The entropy-weighted pseudolabel denoising (EPLD) module dynamically adjusts the learning weights of noisy pixel locations (i.e., pixels with high entropy) to mitigate the adverse influence of noisy pseudolabels on domain adaptation. Nevertheless, the dynamic weighting mechanism of EPLD also results in insufficient contextual learning for target-domain data, especially at locations assigned low weights. To address this issue, we introduce entropy-guided masked image consistency (EMIC), which employs an entropy-guided strategy to mask high-entropy regions and learns spatial contextual relationships in the target domain through masked image consistency, thereby enhancing the model's robustness in visual recognition. Experimental results demonstrate that the proposed method achieves highly competitive performance. The code will be released at https://github.com/woldier/EntSeg**

***Index Terms*—Pseudolabel denoising, remote sensing (RS), self-training (ST), semantic segmentation, unsupervised domain adaptation (UDA).**

I. Introduction

REMOTE sensing refers to a technical methodology that employs telemetry devices mounted on platforms such as artificial satellites, aircraft, and uncrewed aerial vehicles (UAVs) to acquire electromagnetic information from ground objects at a distance, thereby enabling large-scale observation of the Earth's surface. With the continuous advancement of remote sensing (RS) technologies and the emergence of high-resolution Earth observation systems, various satellites, such as Landsat, SPOT, and MODIS, have played a vital role in the development and application of RS imagery. Driven by the rapid progress of deep learning, its exceptional capability for extracting visual features has enabled deep models to demonstrate remarkable potential in RS analysis, becoming the mainstream technique for intelligent interpretation of RS images (RSIs). Deep learning methods have been widely applied in RS scene classification [1], [2], [3], object detection [4], [5], [6], and semantic segmentation [7], [8], [9], [10].

Currently, semantic segmentation of RSIs is a primary research focus. This task aims to use deep learning models to extract reliable semantic information from each pixel in an image, thereby achieving accurate categorization of different land-cover objects. Semantic segmentation has a wide range of real-world applications, providing reliable decision-making support for land-cover mapping, environmental surveillance, and urban planning. With the rapid development of deep learning, numerous high-performing segmentation models have emerged. These methods can be broadly categorized into fully convolutional network (FCN)-based architectures [11], [12], [13] and Transformer-based architectures [14], [15], [16].

In the early stages, supervised learning-based segmentation of RSIs relied heavily on annotated datasets to train deep neural networks, enabling them to learn task-specific features and achieve accurate segmentation. By feeding labeled training samples into the network, the model gradually learns discriminative patterns. However, supervised segmentation methods typically require large-scale, high-quality annotations, significantly increasing human and time costs during data preparation. Consequently, in cross-domain applications (i.e., real-world deployment), supervised segmentation

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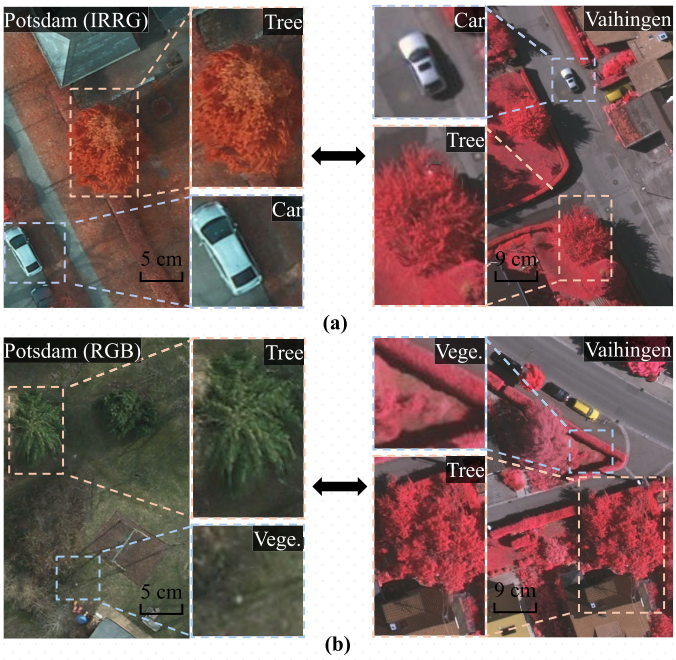


Fig. 1. Differences in data distributions across datasets. Dashed boxes highlight discrepancies of the same category across domains, illustrating their impact on segmentation performance. The scale in the figure indicates the GSD. (a) Different GSD. (b) RS sensor variation.

models often exhibit poor generalization. When the deployment environment (test set) differs from the training domain in geographic conditions or imaging configurations, the mismatch in data distributions leads to sharp performance degradation. This domain shift phenomenon causes deep learning models to deteriorate considerably when handling cross-domain RSIs.

As shown in Fig. 1, the major challenges in cross-domain RSIs' semantic segmentation arise from domain shift caused by variations in ground sampling distance (GSD) and sensor type. These differences affect model generalization capabilities across datasets. From top to bottom, the figure compares GSD and sensor differences. Domain shift is severe in practice due to variations in imaging environment and sensor type across RSIs datasets, resulting in weak generalization of existing methods. A common approach is to first train a model on source-domain data, then fine-tune it on target-domain data via supervised learning. However, annotation of target-domain samples remains labor-intensive and time consuming, severely limiting the deployability of deep learning in operational RS systems.

To address this issue, UDA has been proposed to mitigate the domain shift between a labeled source domain and an unlabeled target domain. Existing UDA methods can be grouped into two categories [17]: adversarial learning-based methods and self-training (ST)-based methods. Adversarial learning methods employ discriminators to align the distributions of source and target data in pixel, feature, or prediction space. However, adversarial training is typically unstable and prone to mode collapse [18], [19], [20]. On the contrary, ST methods transfer knowledge by generating pseudolabels for target images [21], [22], [23], [24], without relying on extra adversarial networks. Nevertheless, due to the domain shift, pseudolabels often contain noise, and learning from these erroneous labels limits model performance. Some methods [25] attempt to mitigate this issue using class prototypes. However, prototype quality may be unreliable for classes with fewer samples. Generally, domain shift increases the uncertainty of predictions on unlabeled target-domain data, leading to high entropy. For example, when the model lacks confidence, it assigns similar probabilities to multiple categories, resulting in high-entropy predictions. If ground-truth labels were available, supervised learning would naturally minimize entropy by increasing model confidence in the correct class. However, pseudolabels, in ST, act as a self-supervised signal; incorrect pseudolabels misguide the model, pulling predictions toward wrong categories.

To address these challenges, we propose EntSeg, comprising two core modules: EPLD and EMIC. The entropy-weighted pseudolabel denoising (EPLD) module uses the per-pixel entropy of pseudolabels as learning weights; high-entropy pixels are more likely to be erroneous and should receive lower weights, thereby reducing the negative impact of noisy pseudolabels. Although EPLD effectively alleviates noise-related issues, the weighting strategy inevitably reduces the learning strength for certain regions, resulting in insufficient modeling of contextual relationships in target-domain images. To compensate for this limitation, we introduce entropy-guided masked image consistency (EMIC). EMIC masks high-entropy regions using an entropy-guided strategy and enforces consistency between predictions generated from the masked target image and those obtained from the full image via an exponential moving average (EMA) teacher. This encourages the model to capture robust spatial context, enabling it to infer reliable predictions for high-entropy areas and addressing the insufficient learning caused by EPLD. By jointly integrating EPLD and EMIC, the proposed EntSeg framework achieves competitive performance. In particular, EntSeg achieves 72.07% mIoU on the Potsdam IRRG → Vaihingen IRRG task and 63.80% mIoU on the Potsdam RGB → Vaihingen IRRG task, outperforming existing state-of-the-art methods.

The contributions of this article are as follows.

1) We propose EntSeg, an entropy-guided learning framework for cross-domain RS semantic segmentation. The framework explicitly addresses the challenges of noisy pseudolabels and insufficient learning in uncertain target regions caused by domain discrepancies.
2) We introduce an EPLD module that dynamically evaluates prediction entropy to assess pseudolabel reliability. By assigning lower weights to high-entropy regions, EPLD suppresses the influence of noisy pseudolabels during ST and improves the stability of target-domain adaptation.
3) We propose an EMIC model that performs targeted masking on high-entropy regions and enforces prediction consistency between masked and full images. This mechanism encourages the model to infer unreliable regions from contextual information, thereby enhancing representation learning for difficult target areas.

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II. Related Work

A. RS Semantic Segmentation

Semantic segmentation of RSIs remains a core yet technically demanding problem due to complex spatial structures and large intraclass variability. A variety of deep neural network architectures have been developed to enhance performance for this task. For example, Mou et al. [26] introduced dedicated spatial and channel relationship modeling components to explicitly capture dependencies among features across different spatial locations and feature channels, thereby strengthening contextual interaction. Li et al. [27] developed a hierarchical wavelet feature enhancement module that integrates multiresolution wavelet decomposition into the feature extraction pipeline, enabling more effective characterization of fine-grained structures in RSIs. Yang et al. [28] propose a novel online spectral information compensation network (OSICN) for hyperspectral image classification.

Given the substantial reliance of fully supervised frameworks on densely annotated data, recent efforts have increasingly shifted toward weakly supervised RS segmentation. Miao et al. [29] proposed EACE, a framework comprising a class activation network and an edge-aware network to produce high-confidence pseudolabels. The method first learns from limited annotated samples and subsequently exploits pseudolabel supervision to progressively improve segmentation performance. To address the shortage of accurate pixel-level annotations that constrains representation learning, Lu et al. [30] presented a weakly supervised approach that incorporates self-supervised pretraining and pseudolabel denoising strategies to alleviate label ambiguity and noise accumulation. More recently, vision Transformer (ViT)-based architectures have been incorporated into weakly supervised segmentation pipelines. However, their global attention mechanism may lead to oversmoothing in densely distributed scenes, weakening boundary discrimination. To mitigate this issue, Zhang et al. [31] introduced ATSG, which integrates two specialized components: an adaptive token linking (Adalink) module and a segment anything model-guided boundary refiner (SGBR). Adalink enhances token interactions to suppress spurious activations in cluttered regions and improve pseudolabel reliability. The SGBR module leverages the zero-shot segmentation capability of the segment anything model (SAM) to refine object boundaries, resulting in more complete delineation of small targets and improved overall segmentation accuracy.

B. UDA Semantic Segmentation

UDA for semantic segmentation aims to adapt a model trained on labeled source-domain data to a target domain without requiring target-domain annotations. Recently, many remarkable works have made huge progress in applying UDA to cross-domain semantic segmentation tasks.

Adversarial learning is a commonly adopted strategy in many high-performing UDA methods. Tsai et al. [32] observed that the source and target domains share strong similarities in semantic layout and therefore constructed a multilevel adversarial network to exploit structural consistency across the output spaces of different domains. Vu et al. [33] introduced a discriminator to encourage the entropy distribution of the target domain to approximate that of the source domain. However, discriminator networks are often highly sensitive to hyperparameters, making it challenging for models to learn similar feature distributions across domains. In addition, adversarial learning itself can lead to unstable training, further increasing optimization difficulty.

ST, a typical nonadversarial UDA paradigm, has received increasing attention in cross-domain semantic segmentation. Zou et al. [34] first introduced an ST method for UDA semantic segmentation by treating pseudolabels as discrete latent variables and optimizing a unified learning objective. Tranheden et al. [22], Zhou et al. [35], Hoyer et al. [23], and Chen et al. [36] enhanced domain adaptation by generating pseudolabels with improved reliability, consistency, and class balance.

C. UDA Semantic Segmentation in RS

Although cross-domain semantic segmentation of RSIs has not been thoroughly investigated, numerous strong studies have emerged in the RS community.

Benjdira et al. [37] were the first to address the domain adaptation problem for RS semantic segmentation, employing generative adversarial networks (GANs) to mitigate domain shift and achieving promising results. Following this line of research, Li et al. [38] introduced DualGAN into UDA-based RS semantic segmentation and demonstrated its effectiveness in cross-domain RS scenarios. Cai et al. [39] proposed a bidirectional adversarial learning framework that enforces bidirectional semantic consistency for RSI segmentation. Zhou et al. [40] introduced a UDA semantic segmentation framework that integrates RSIs with a digital surface model (DSM) using a multipath encoder and multitask decoder while aligning global cross-domain data distributions. Wang et al. [9] further incorporated contrastive learning to alleviate insufficient learning on target-domain samples. However, these ST approaches do not adequately address the adverse effects of noisy pseudolabels, which can mislead the model and impair adaptation performance. Different from [9], which focuses on contrastive representation learning, this work investigates UDA segmentation from the perspective of pseudolabel reliability.

To alleviate the adverse effects caused by noisy pseudolabels, Yan et al. [41] incorporated an average-teacher model into the UDA paradigm and developed a cross-average teacher network, where reliable pseudolabels are generated for target-domain data and global cross-domain distribution alignment is achieved. On the contrary, this work adopts a different strategy. Instead of relying on teacher-based reliability filtering alone, we dynamically reweight pseudolabels during training to reduce the learning emphasis on noisy pixel locations. Since reliable pseudolabel predictions typically exhibit low entropy, we assign entropy-based weights to pseudolabels to suppress the influence of noisy annotations and mitigate their negative impact on the model.

In addition to RS semantic segmentation, recent studies have also explored cross-domain learning for other RS tasks, such as hyperspectral image classification and change detection. For example, Liu et al. [42] proposed a dual classification head

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self-training network (DHSNet) for cross-domain hyperspectral image classification. To better integrate high-dimensional spectral information, Liu et al. [43] further introduced an adaptive smooth adversarial learning (ASAL) method for cross-domain hyperspectral image classification. In the area of change detection, Zhou et al. [44] proposed a temporal self-construction cross-domain learning (TSCCD) framework for domain adaptive hyperspectral image change detection.

D. Masked Image Modeling

In natural language processing, predicting masked portions of an input sequence is a powerful self-supervised pretraining task [45], [46]. Recently, this idea has been successfully transferred to computer vision as masked image modeling, where a model receives a partially masked image and attempts to reconstruct the missing regions, such as variational autoencoders (VAEs) [47] features, histograms of oriented gradients (HOGs) [48] features, or color information. To investigate how different masking strategies affect model learning, several approaches have been proposed, including block-wise masking [47], random patch masking [49], [50], and attention-guided masking [51], [52].

In the context of UDA, Hoyer et al. [53] introduced image masking for consistency representation learning, where random regions of a target image are masked to facilitate the learning of contextual relationships. However, random masking does not distinguish between easy and difficult regions (e.g., visually ambiguous areas with high entropy). Differentiating these regions and focusing on the recovery of the difficult ones is beneficial for learning richer spatial contextual information. Moreover, EPLD inevitably leads to insufficient and unreliable learning in noisy regions. Therefore, by placing greater emphasis on recovering contextual information for high-entropy regions, our method compensates for the shortcomings associated with insufficient or unreliable learning and strengthens the model's overall representational robustness.

III. Method

A. Preliminary

To simplify the problem, we assume that the source-domain and target-domain images share the same spatial resolution, each with dimensions $H \times W \times 3$, where H and W denote the image height and width, and 3 represents the number of channels. In addition, both domains contain the same number of semantic categories. Given an image x_S^i and its corresponding label y_S^i, the source domain is defined as follows:

$$D_S = \left\{ \left(x_S^i, y_S^i\right) \,\middle|\, x_S^i \in \mathbb{R}^{H\times W\times 3},\ y_S^i \in \mathbb{R}^{H\times W\times C} \right\}_{i=1}^{N_S} \tag{1}$$

where C denotes the number of categories. Similarly, the target domain is defined as follows:

$$D_T = \left\{ x_T^i \,\middle|\, x_T^i \in \mathbb{R}^{H\times W\times 3} \right\}_{i=1}^{N_T}. \tag{2}$$

Due to the unsupervised nature of the problem, only the images x_T^i are accessible while the labels y_T^i are unavailable. Subscripts S and T denote the source and target domains, respectively, and N_S and N_T represent the number of samples in each domain. The source-domain label $y_S^{i,h,w}$ at spatial location (h, w) is expressed as a one-hot vector of length C, where $h \in [1, \ldots, H]$ and $w \in [1, \ldots, W]$.

B. EntSeg Overview

Fig. 2 presents an overview of the training workflow of **EntSeg**. The model g_θ denotes a deep learning architecture designed for semantic segmentation, consisting of a backbone network f and a segmentation classification head, where θ represents the learnable parameters of g_θ. To ensure more reliable pseudolabels during ST, we adopt an EMA-based teacher model t_θ, which generates smoother and more stable predictions for supervising the student model g_θ. The teacher network t_θ shares the same architecture as g_θ. However, its parameters are not updated via gradient backpropagation due to the use of *stop gradient* and are instead updated at the end of each iteration via EMA with the weights of g_θ.

During each training iteration, the proposed method jointly processes source-domain images x_S^i and target-domain images x_T^i. It is important to note that only the labels of the source domain y_S^i are accessible in this process. The source-domain pairs (x_S^i, y_S^i) are used to train g_θ via supervised learning. Meanwhile, the target-domain image x_T^i is first forwarded through the teacher model t_θ to produce a pseudolabel p_T^i and the corresponding entropy map e_T^i. Subsequently, on the one hand, the prediction of g_θ on x_T^i, together with its entropy, is used to optimize the parameters of g_θ through the proposed EPLD module, thereby mitigating the impact of noisy pseudolabels. On the other hand, the entropy map is used to mask high-entropy regions of x_T^i, producing a partially occluded masked image $x_{T,\text{mask}}^i$. The prediction generated by g_θ on this masked image is then used to optimize the model via the proposed EMIC module, which improves the learning of spatial contextual relationships in the target domain. After each iteration, the teacher model t_θ is updated using an EMA of the parameters of the current student model g_θ. The EPLD and EMIC modules will be discussed in detail in Sections III-F and III-G, respectively.

In our framework, the student network and teacher network share the same architecture and are initialized with the same weights at the beginning of training. During training, the student network is updated using gradient-based optimization during each iteration, while the teacher network is updated using an EMA of the student parameters after each iteration.

C. Supervised Learning on the Source Domain

If domain adaptation is not considered, the network g_θ can be trained solely on the source domain using the cross-entropy loss. The loss function is given as follows:

$$\mathcal{L}_S^i = -\sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{c=1}^{C} y_S^{i,h,w,c} \cdot \log g_\theta\left(x_S^i\right)^{h,w,c} \tag{3}$$

where $g_\theta(x_S^i)$ denotes the per-pixel prediction of the source-domain image x_S^i, where each spatial position (h, w) corresponds to a vector of class probabilities. Due to the presence of domain gaps, training only on the source domain typically

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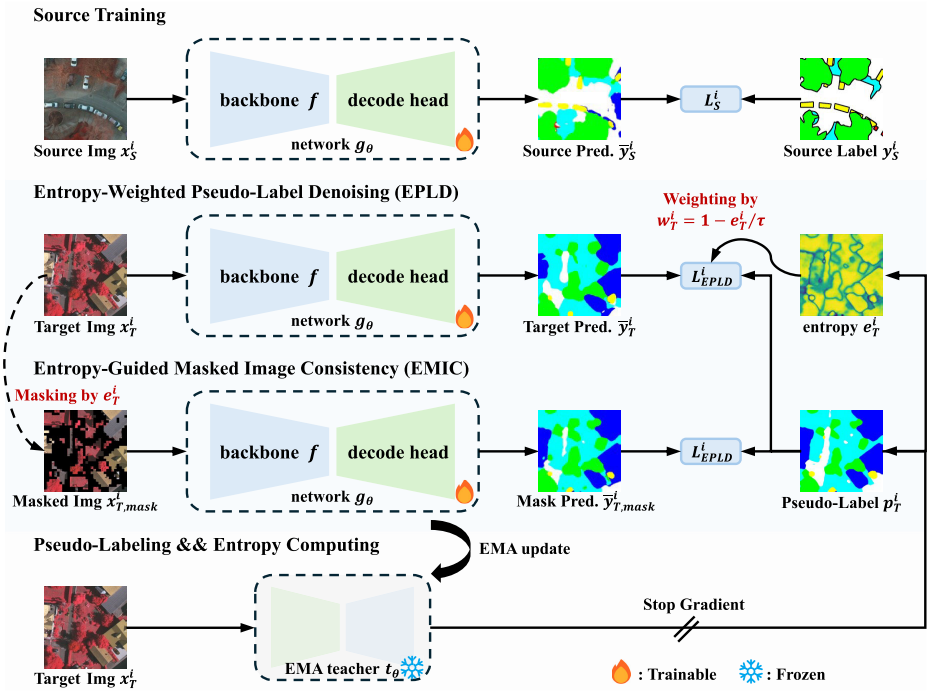


Fig. 2. UDA method EntSeg with proposed EPLD and EMIC. In UDA, a network is typically trained with a supervised loss on the source domain and an unsupervised adaptation loss with pseudolabels that are generated based on the complete image by an EMA teacher on the target domain. EPLD weighted loss to ignore noisy pseudolabels. EMIC enforces the consistency between predictions of entropy-guided masked target images and pseudolabels.

yields suboptimal performance on the target domain, as the network fails to generalize well to unseen target-domain samples.

D. EMA Teacher Model

Pseudolabels p_T^i can be generated either online or offline [9]. In online ST, the teacher model t_θ is updated together with the student model g_θ during training. The teacher and student models share identical architectures, and gradient flow to the teacher network is stopped to prevent backpropagation through t_θ.

To mitigate the parameter perturbation caused by the domain gap between source and target domains, the teacher model parameters are updated at each iteration via the EMA, which stabilizes pseudolabel generation. The update rule is expressed as follows:

$$S_{t+1}(t_\theta) \leftarrow \alpha \cdot S_t(t_\theta) + (1-\alpha) \cdot S_t(g_\theta) \tag{4}$$

where $S_t(\cdot)$ denotes the model parameters at training step t and $\alpha \in [0,1]$ controls the contribution of the previous teacher parameters. This update strategy not only improves pseudolabel reliability but also alleviates the impact of domain discrepancies, thereby enhancing the network's generalization capability on the target domain.

E. Pseudolabel Generation

To transfer knowledge from the source domain to the target domain, the teacher model t_θ produces pseudolabels for unlabeled target-domain samples. The pseudolabel of pixel (h,w) is defined as follows:

$$p_T^{i,h,w,c} = \left[c = \arg\max_{c'} t_\theta\left(x_T^i\right)^{h,w,c'} \right] \tag{5}$$

where $[\cdot]$ denotes the Iverson bracket. The output $t_\theta(x_T^i)$ represents the per-pixel class prediction for the target-domain image x_T^i. The teacher model parameters remain fixed during each iteration and are updated only via EMA.

F. Entropy-Weighted Pseudolabel Denoising

Due to domain shift, pseudolabels often contain noise arising from incorrect predictions. Learning from such noisy pseudolabels can impair model performance. In general, predictions on target-domain images tend to exhibit high uncertainty and thus high entropy. For example, when the model is unsure about the target data, it assigns similar probabilities to multiple categories, resulting in large entropy. In particular, EPLD aims to reduce the negative impact of noisy pseudolabels. Since predictions with high entropy are more likely to be incorrect, EPLD assigns lower weights to these uncertain pixels and higher weights to more confident ones. In this way, the model focuses more on reliable pseudolabels while suppressing the influence of noisy supervision during training.

Because lower entropy indicates higher confidence, entropy can serve as a measure of pseudolabel reliability: high-confidence (low-entropy) labels should receive larger weights, while low-confidence (high-entropy) labels should be down-weighted. First, the predicted logits are passed through the softmax function

$$\text{prob}_T^{i,h,w,c} = \underset{c'}{\text{softmax}}\, t_\theta\left(x_T^i\right)^{h,w,c'}. \tag{6}$$

The entropy at each pixel is computed as follows:

$$e_T^{i,h,w} = \sum_c \text{prob}_T^{i,h,w,c} \cdot \log\left(\text{prob}_T^{i,h,w,c} + \varepsilon\right) \tag{7}$$

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where $\varepsilon = 1 \times 10^{-6}$, which is a small constant for numerical stability. The reliability weight is then defined as follows:

$$w_T^{i,h,w} = 1 - \frac{e_T^{i,h,w}}{\tau}, \qquad \tau = \log(C) \tag{8}$$

which normalizes entropy into $[0, 1]$. The target-domain loss is defined as follows:

$$\mathcal{L}_{\text{EPLD}}^i = -\sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{c=1}^{C} w_T^{i,h,w} \cdot p_T^{i,h,w,c} \cdot \log g_\theta \left(x_T^i\right)^{h,w,c}. \tag{9}$$

By weighting pseudolabels with entropy, EPLD suppresses the negative impact of noisy pseudolabels and alleviates performance degradation caused by label noise.

G. Entropy-Guided Masked Image Consistency

While EPLD reduces the influence of noisy pseudolabels, it also decreases the contribution of high-entropy regions during training (where the weights are small), resulting in insufficient learning on the target domain. This limits the network's ability to capture contextual relationships. To address this issue, we propose a masking mechanism that masks high-entropy regions and enforces consistency between the predictions of masked and unmasked images. In particular, EMIC masks high-entropy areas in the input image and encourages the model to infer these regions using surrounding contextual information. By enforcing consistency between masked and unmasked predictions, the model learns stronger spatial relationships and produces more reliable segmentation results.

The entropy map e_T^i computed via (7) is reshaped into a 1-D vector and sorted in descending order

$$\sigma_\downarrow : 1, 2, \ldots, n, \quad n = H \times W \tag{10}$$

such that $e_{T,\sigma_\downarrow(\rho)}^i \geq e_{T,\sigma_\downarrow(\upsilon)}^i$ for $\rho < \upsilon$. The top-k high-entropy pixels are selected

$$\text{idx}_{\text{mask}} = \left\{\sigma_\downarrow(1), \ldots, \sigma_\downarrow(k)\right\}, \quad k = \gamma \cdot n \tag{11}$$

where $\gamma \in [0, 1]$ denotes the masking ratio. The mask is defined as follows:

$$M_{\text{ent}}^i = \begin{cases} 0, & \text{if } i \in \text{idx}_{\text{mask}} \\ 1, & \text{otherwise.} \end{cases} \tag{12}$$

The masked image is obtained via element-wise multiplication

$$x_{T,\text{mask}}^i = M_{\text{ent}}^i \odot x_T^i. \tag{13}$$

The EMIC loss, enforcing consistency between the predictions of the masked image and the pseudolabels generated by the EMA teacher, is expressed as follows:

$$\mathcal{L}_{\text{EMIC}}^i = -\sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{c=1}^{C} w_T^{i,h,w} \cdot p_T^{i,h,w,c} \cdot \log g_\theta \left(x_{T,\text{mask}}^i\right)^{h,w,c}. \tag{14}$$

EMIC enhances the consistency between the predictions of entropy-guided masked target images and the predictions of the full-image pseudolabels generated by the EMA teacher model, thereby improving contextual understanding and strengthening target-domain learning.

IV. Experimental Results and Analysis

A. Datasets

To evaluate the effectiveness of the proposed method on cross-domain RS segmentation, we adopt two benchmark datasets: Potsdam and Vaihingen. The Potsdam and Vaihingen datasets are part of the ISPRS 2-D semantic segmentation benchmark [54]. The Potsdam (POT) dataset contains 38 aerial images with a spatial resolution of 6000×6000 and a GSD of 5 cm. It provides three imaging modalities: IRRG, RGB, and RGBIR, where the first two modalities contain three channels and the last one contains four channels. In this study, we use the IRRG and RGB modalities. The Vaihingen (VAI) dataset provides 33 aerial images with resolutions ranging from 1996×1996 to 3816×2550 and a GSD of 9 cm. It contains a single modality: IRRG. Both datasets share the same six semantic classes. It is worth noting that objects may appear visually different in the IRRG modality (e.g., trees and low vegetation appear red).

Following previous works [9], [20], we crop the images into smaller 512×512 patches to reduce computational cost. For POT and VAI, cropping strides are set to 512 and 256, respectively. The resulting patches are split into training and testing sets.

Because POT mainly contains urban areas with many vehicles, while VAI includes more residential areas and farmland, their visual appearances differ significantly despite having identical label spaces. These characteristics make the two datasets suitable for evaluating UDA methods. Based on these properties, we construct two cross-domain RS semantic segmentation tasks.

1) Potsdam IRRG to Vaihingen IRRG (POT IRRG → VAI IRRG).
2) Potsdam RGB to Vaihingen IRRG (POT RGB → VAI IRRG).

B. Evaluation Metric

Following prior UDA research in RS semantic segmentation, we adopt the $F1$-score and the intersection over union (IoU) as evaluation metrics. The metrics are computed as follows:

$$\text{IoU} = \frac{TP}{TP + FP + FN} \tag{15}$$

where TP, FP, and FN denote true positives, false positives, and false negatives, respectively. These metrics provide a reliable assessment of model performance, particularly under class imbalance and allow us to systematically compare different methods in cross-domain RS segmentation tasks.

C. Implementation Details

1) Augmentation: We adopt the default data processing pipeline, which includes randomly applied resizing, cropping, and flipping. Following DACS [9], [22], we additionally introduce color jitter, Gaussian blur, and ClassMix [55] to enhance data diversity and improve feature robustness.

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2) Network Architecture: Given the strong performance of the Transformer-based SegFormer [15], we employ mix Transformers (MiT) [15], that pretrained on ImageNet [56], as the backbone f. Following previous UDA works [9], [23], we adopt MiT-B5 [15] that was pretrained on ImageNet [56], as the backbone f to ensure fair comparison with existing methods. MiT-B5 is a hierarchical Transformer consisting of four stages with depths [3, 6, 40, and 3] and feature channel dimensions [64, 128, 320, and 512], respectively. Each stage begins with an overlapped patch embedding layer implemented with a convolutional kernel of size 7×7 (stride 4) in the first stage and 3×3 (stride 2) in subsequent stages, followed by multiple Transformer encoder blocks. The backbone produces a multiscale feature pyramid with resolutions $H/4 \times W/4$, $H/8 \times W/8$, $H/16 \times W/16$, and $H/32 \times W/32$. Unlike CNN-based models such as DeepLab [57], MiT-B5 enables more effective modeling of global and local contextual information. For multiscale feature fusion, we use the context-aware decoder head introduced in [23], which has demonstrated excellent performance in UDA semantic segmentation. The decoder projects each feature map to 256 channels and upsamples them to a unified spatial resolution of $H/4 \times W/4$. These features are fused and processed by depth-wise dilated separable convolutions with dilation rates of 1, 6, 12, and 18 to capture multiscale contextual information.

3) Training: For the network g_θ, we use the AdamW optimizer with $\beta = (0.9, 0.999)$ and weight decay 0.01. The learning rate for the backbone is set to 6×10^{-4}, while the decoder head uses 6×10^{-5}. A linear warm-up strategy is applied for the first 1500 iterations, followed by poly decay with a power of 1.0. The decay coefficient for the EMA teacher network t_θ is set to $\alpha = 0.99$. The masking ratio is $\gamma = 0.5$, and the patch size is 16. Following the standard MMSegmentation training protocol, optimization is performed in an iteration-based manner rather than an epoch-based manner. The total number of training iterations is 40 000, and the batch size is 12 (six from the source domain and six from the target domain).

Our implementation is based on PyTorch [58] and MMSegmentation[1] [59]. All baselines are reproduced under the MMSegmentation framework for fair comparison. All experiments are conducted on an NVIDIA A100 GPU.

D. Quantitative Comparison With Existing UDA Methods

In this section, we compare the proposed method with several representative approaches on two cross-domain RS segmentation tasks: POT IRRG → VAI IRRG and POT RGB → VAI IRRG. Several competitive and widely adopted baselines, including AdaptSegNet [32], ADVENT [33], CaGAN [60], CIA-UDA [61], Zhu et al. [62], FSGA-CB [63], AACDN [64], DACS [22], DAFormer [23], ST-DASegNet [20], SiamSeg [9], Geng et al. [65], and GRLDASeg [66], are included in our evaluation.

1) POT IRRG → VAI IRRG: Table I reports the results on the POT IRRG → VAI IRRG cross-domain semantic segmentation task. AdaptSegNet achieves an mIoU of 48.13%, showing relatively poor performance. This is primarily because the method relies on adversarial learning to align feature distributions between the source and target domains. While distribution alignment can reduce domain shift, adversarial training often leads to the loss of domain-specific information. This limitation is particularly evident for RS data containing strong spatial and spectral differences, resulting in reduced adaptation ability. On the contrary, ADVENT enhances the model's prediction capability on target-domain data by minimizing entropy, slightly improving performance to 49.50%. CaGAN adopts a class-aware strategy and further improves the performance to 51.27% mIoU. CIA-UDA leverages category-level information to enhance adaptation performance, achieving 55.21% mIoU. Zhu et al. [62] utilize domain-invariant prototypes to obtain improved adaptation results, reaching 56.04% mIoU. FSGA-GB achieved a 46.82% mIoU thanks to the proposed Self-AdaIN module, which precisely controls the style transformation process, and a class balance loss function. AACDN, by comparing low- and high-level features, achieved a 56.14% mIoU.

DACS combines target-domain data augmentation with pseudolabel-based training and achieves competitive adaptation performance, reaching an mIoU of 60.20%. DAFormer further improves performance by introducing a decoder specifically designed for UDA, resulting in an mIoU of 63.25%. ST-DASegNet combines adversarial learning with ST and achieves an mIoU of 64.33%, indicating that integrating these two training strategies is an effective solution. SiamSeg exploits contrastive learning to strengthen target-domain representation learning and achieves 65.55% mIoU, demonstrating that contrastive objectives are beneficial for cross-domain segmentation. Geng et al. [65] leverage multiscale contrastive domain adaptation and obtain an mIoU of 62.17%. GRLDASeg exploits general representation learning and achieves highly competitive performance, reaching an mIoU of 67.89%. However, despite their advancements, these approaches still rely on simple thresholding for pseudolabel generation and do not account for the detrimental effect of noisy pseudolabels. Such noise introduces erroneous supervision and limits the model's adaptation capability.

To address this issue, the proposed EntSeg method integrates EPLD and EMIC and achieves substantial performance gains. EPLD employs entropy-weighted learning to reduce the influence of noisy pseudolabels, effectively suppressing noise-induced errors. Meanwhile, EMIC masks unreliable regions using entropy and encourages the network to infer accurate predictions for masked areas by leveraging contextual information. As a result, EntSeg achieves the highest mIoU of 72.07%, significantly outperforming all existing methods.

2) POT RGB → VAI IRRG: Table I presents the results on the POT RGB → VAI IRRG task. AdaptSegNet obtains an mIoU of only 38.40%, exhibiting poor performance similar to the results reported in task POT IRRG → VAI IRRG. The mIoU of ADVENT increases slightly to 39.03%. Subsequent methods further improved the performance of adversarial learning-based approaches. In particular, CaGAN, CIA-UDA, and the method proposed by Zhu et al. [62] achieved mIoU

[1] https://github.com/open-mmlab/mmsegmentation

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TABLE I

CROSS-DOMAIN RS SEGMENTATION RESULTS ON TASKS POT IRRG → VAI IRRG AND POT RGB → VAI IRRG. BOLD VALUES INDICATE THE BEST PERFORMANCE. THE ABBREVIATION LOW VEG. IN THE TABLE STANDS FOR LOW VEGETATION, WHILE IMP. SURF. STANDS FOR IMPERVIOUS SURFACE

(a) POT IRRG → VAI IRRG

Method	Year	Clutter	Car	Tree	Low Veg.	Building	Imp. Surf.	mIoU (%)
Adversarial Learning UDA								
AdaptSegNet [32]	CVPR'18	16.35	17.38	66.93	45.34	76.79	66.01	48.13
ADVENT [33]	CVPR'19	10.69	31.41	66.68	45.86	75.04	67.32	49.50
CaGAN [60]	TGRS'20	25.94	49.36	63.84	42.41	66.63	59.40	51.27
CIA-UDA [61]	TGRS'23	27.80	52.91	64.11	48.03	75.13	63.28	55.21
Zhu et al. [62]	TGRS'23	33.60	43.97	64.63	51.76	76.13	66.18	56.04
FSGA-CB [63]	JSTARS'25	13.16	**58.18**	66.81	50.83	86.77	64.97	56.78
AACDN [64]	JSTARS'26	10.19	50.13	65.23	50.78	85.89	74.66	56.14
Self-Training UDA								
DACS [22]	CVPR'21	33.96	48.73	66.75	51.10	85.65	75.00	60.20
DAFormer [23]	CVPR'22	17.83	50.33	74.39	66.26	90.09	80.60	63.25
ST-DASegNet [20]	JAG'24	**67.03**	43.38	67.36	48.57	85.23	74.43	64.33
SiamSeg [9]	Arxiv'25	27.60	52.51	77.69	65.73	89.00	80.75	65.55
Geng et al. [65]	TGRS'25	49.37	52.49	67.38	47.27	82.90	73.62	62.17
GRLDASeg [66]	TGRS'25	50.02	57.52	74.12	54.42	88.65	82.62	67.89
EntSeg (ours)	-	51.77	54.01	**80.81**	**71.39**	**91.33**	**83.12**	**72.07**

(b) POT RGB → VAI IRRG

Method	Year	Clutter	Car	Tree	Low Veg.	Building	Imp. Surf.	mIoU (%)
Adversarial Learning UDA								
AdaptSegNet [32]	CVPR'18	12.49	19.94	61.37	14.26	64.76	57.58	38.40
ADVENT [33]	CVPR'19	4.30	38.62	60.50	6.17	67.03	57.59	39.03
CaGAN [60]	TGRS'20	11.63	47.30	55.46	31.52	70.44	55.74	45.35
CIA-UDA [61]	TGRS'23	13.50	52.28	63.43	33.31	79.71	62.63	50.81
Zhu et al. [62]	TGRS'23	21.92	41.19	53.01	40.11	68.08	56.97	46.88
FSGA-CB [63]	JSTARS'25	12.67	46.01	34.25	46.89	77.24	63.86	46.82
AACDN [64]	JSTARS'26	6.87	44.51	56.01	30.10	79.92	60.21	46.27
Self-Training UDA								
DACS [22]	CVPR'21	18.29	43.04	33.81	35.05	82.50	61.04	45.62
DAFormer [23]	CVPR'22	12.97	40.69	65.61	54.30	83.42	72.67	54.94
ST-DASegNet [20]	JAG'24	**36.03**	43.15	64.65	34.69	**84.09**	68.36	55.16
SiamSeg [9]	Arxiv'25	13.23	51.14	71.09	40.10	81.56	73.15	55.04
Geng et al. [65]	TGRS'25	10.32	**62.59**	67.56	41.46	79.36	61.57	53.81
GRLDASeg [66]	TGRS'25	25.09	59.85	69.54	51.73	85.29	79.12	61.77
EntSeg (ours)	-	15.38	58.49	**79.00**	**59.54**	70.61	**79.80**	**63.80**

scores of 45.35%, 50.81%, and 46.88%, respectively. However, their performance remains limited due to the inherent reliance on adversarial training.

DACS achieves 45.62% mIoU by leveraging pseudolabels and target-domain augmentation. DAFormer further improves the result to 54.94% owing to its UDA-oriented decoder design. SiamSeg reaches 55.04% mIoU through contrastive learning, highlighting enhanced capability in learning from target-domain data. However, all these approaches still suffer from noisy pseudolabels, which can misguide training and introduce significant prediction errors.

On the contrary, EntSeg achieves the best performance on this dataset with an mIoU of 63.80%. EPLD attenuates the negative impact of noisy pseudolabels through entropy-weighted supervision, while EMIC strengthens contextual learning by enforcing entropy-guided masked prediction consistency. These complementary mechanisms enable EntSeg to generate more reliable predictions in complex target-domain regions, demonstrating superior adaptation capability and robustness compared to all baselines.

In summary, EntSeg consistently achieves superior performance on both cross-domain RS semantic segmentation tasks owing to the combined effectiveness of EPLD and EMIC. By further improving pseudolabel noise suppression and exploiting contextual information, EntSeg demonstrates strong adaptability and robustness across diverse and challenging scenarios.

E. Qualitative Comparison With Existing UDA Methods

1) POT IRRG → VAI IRRG: Fig. 3 presents qualitative results for the POT IRRG → VAI IRRG task. Different categories in the images are distinguished by colors, where red denotes clutter, yellow denotes car, green denotes tree, light green denotes low vegetation, blue denotes building, and white denotes impervious surface. The figures illustrate, in order, the original image, the segmentation outputs of AdaptSegNet [32], ADVENT [33], DACS [22], DAFormer [23], SiamSeg [9], the proposed EntSeg, and the ground-truth labels.

By examining the visualized results, we observe that AdaptSegNet and ADVENT exhibit a large number of incorrect segmentations, particularly around complex boundaries. Because these adversarial learning-based methods enforce alignment between source- and target-domain feature distributions, they tend to overalign the domains and consequently

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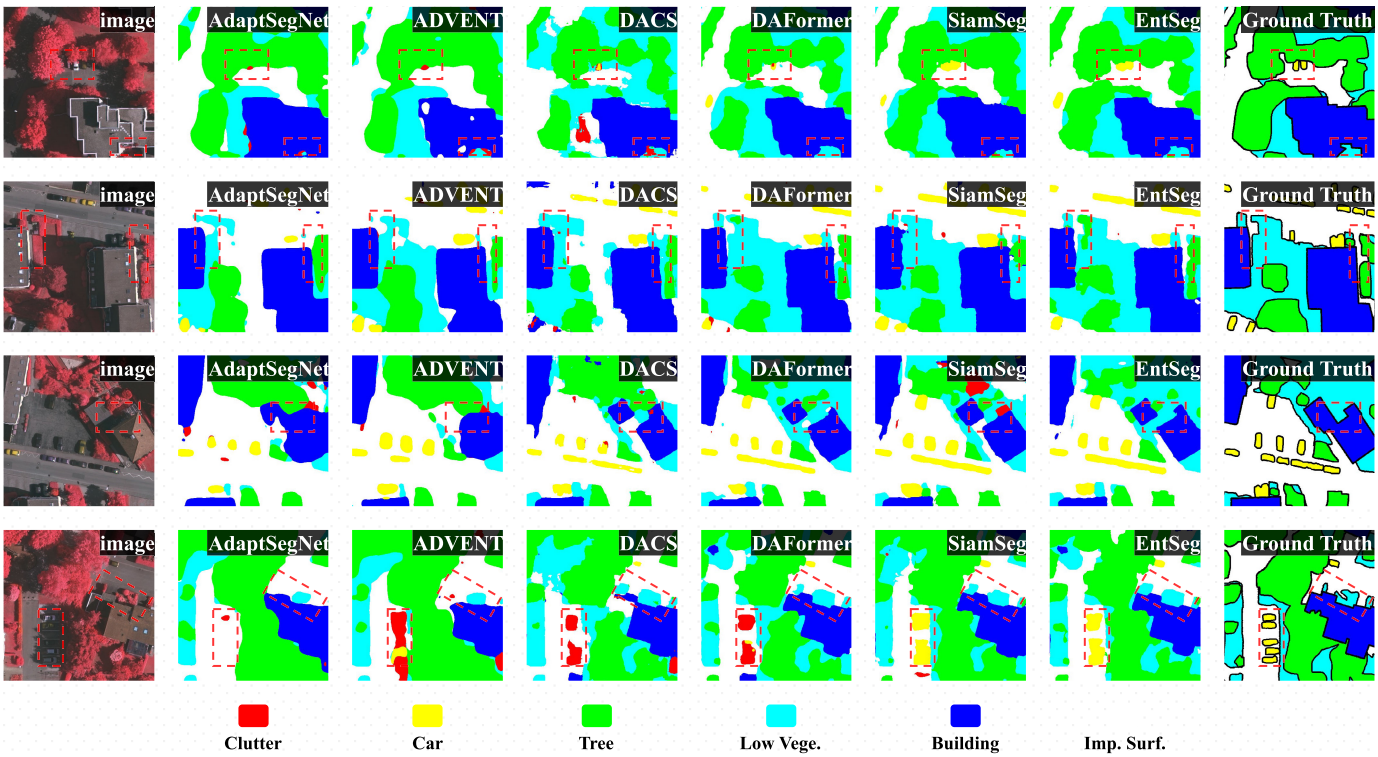


Fig. 3. Visualization of results on cross-domain tasks Potsdam IRGB → Vaihingen IRRG.

lose target-domain-specific characteristics, leading to distorted predictions.

DACS and DAFormer improve segmentation accuracy by incorporating target-domain image augmentations and pseudolabel-based ST. Although these methods achieve better visual quality, their reliance on simple pseudolabel generation makes them vulnerable to noisy pseudolabels, which continue to degrade target-domain precision.

SiamSeg enhances target-domain representation learning via contrastive learning, resulting in more accurate predictions and finer detail preservation. However, it still cannot fully eliminate the adverse effects of noisy pseudolabels, causing inaccurate predictions in several regions.

The proposed EntSeg achieves the most accurate segmentation results. The EPLD module attenuates the influence of noisy pseudolabels during training, while the EMIC module masks high-entropy regions and encourages the network to infer these masked areas from the surrounding context. Together, EPLD and EMIC substantially improve the target-domain adaptability of EntSeg and yield the best segmentation performance among all compared methods.

2) POT IRRG → VAI IRRG: Fig. 4 shows the qualitative results for the POT IRRG → VAI IRRG task, further validating the effectiveness of EntSeg for cross-domain RS semantic segmentation. The category–color mapping remains identical to that in Fig. 3.

The visual comparisons indicate that EntSeg performs considerably more robustly than AdaptSegNet and ADVENT in the target domain. In particular, EntSeg yields more accurate predictions for buildings, impervious surface, and low vegetation, whereas AdaptSegNet and ADVENT produce substantial errors in these categories.

TABLE II

EFFECTIVENESS OF EPLD AND EMIC. A CHECK MARK (✓) INDICATES THE COMPONENT IS USED; A (-) INDICATES IT IS NOT. THE ABERRATION PI→V STANDS FOR POT IRRG → VAI IRRG, WHILE PR→V STANDS FOR POT RGB → VAI IRRG

Method	EPLD	EMIC	Pi→V	Pr→V
Method 1	-	-	63.83	53.27
Method 2	✓	-	66.16	58.47
Method 3	-	✓	65.31	55.43
EntSeg	✓	✓	**72.07**	**63.80**

DACS and DAFormer also demonstrate strong performance in this task, especially for tree and impervious surface. Nevertheless, the presence of noisy pseudolabels continues to affect their accuracy in certain regions. On the contrary, EntSeg benefits from EPLD's ability to mitigate pseudolabel noise and EMIC's ability to leverage contextual consistency, leading to significantly improved segmentation accuracy—especially along object boundaries.

Overall, in the POT IRRG → VAI IRRG, EntSeg surpasses all compared approaches across multiple categories through its innovative pseudolabel denoising and context-aware learning mechanisms, demonstrating superior cross-domain adaptability.

F. Ablation Study

1) Effectiveness of EPLD and EMIC: To validate the effectiveness of EPLD and EMIC, a series of comparative experiments was conducted to assess the contribution of each method to model performance, as summarized in Table II. These experiments were designed to quantify the impact of

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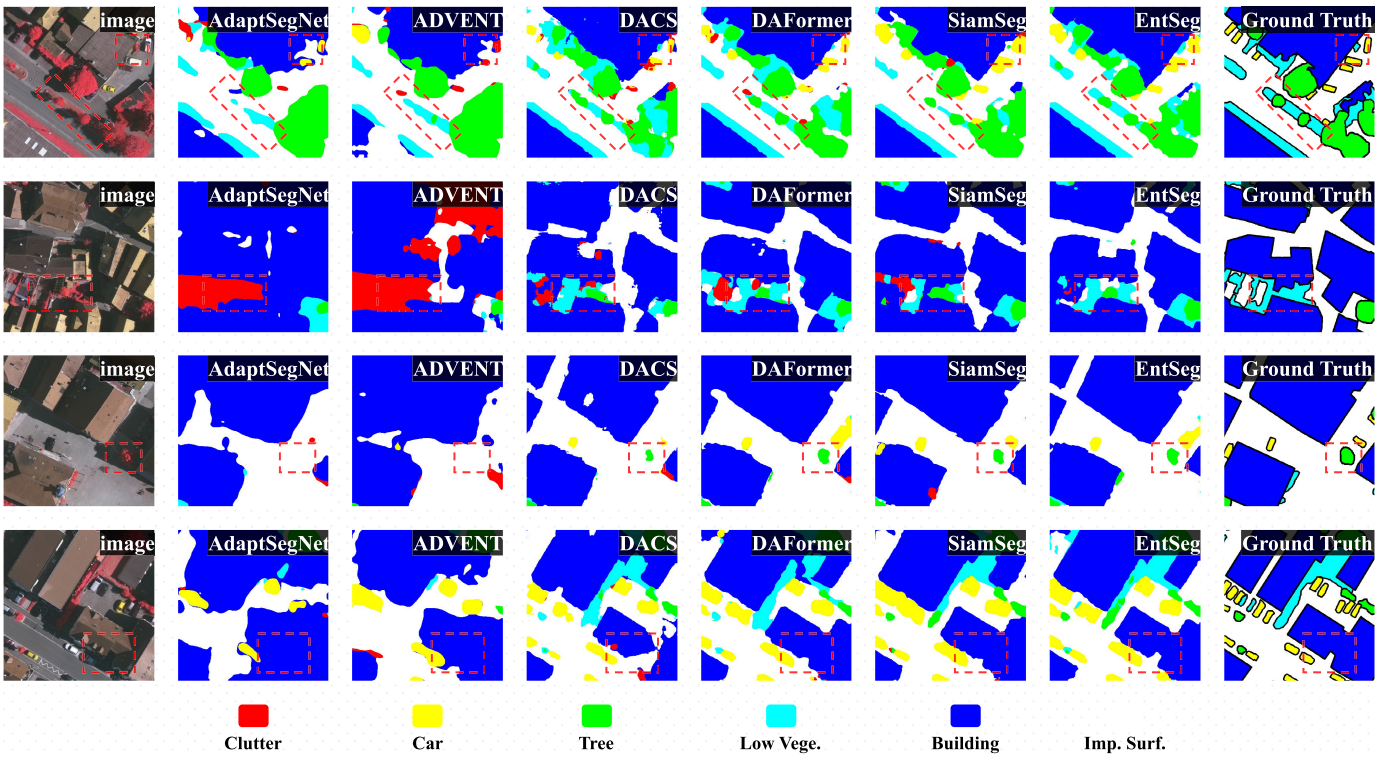


Fig. 4. Visualization of results on cross-domain tasks Potsdam RGB → Vaihingen IRRG.

EPLD and EMIC under different configurations. In the version without EPLD, a globally uniform weight scheme from [22] was used, which is generally susceptible to noisy pseudolabels, thereby degrading model adaptability and prediction accuracy.

The results indicate that without EPLD and EMIC, Method 1 achieves moderate performance with an mIoU of 63.83% and 53.27 on task POT IRRG → VAI IRRG and POT RGB → VAI IRRG, respectively. This demonstrates that in the absence of effective denoising strategies, noisy pseudolabels significantly interfere with model learning, particularly in complex regions where accurate target boundaries and fine details are difficult to predict, leading to reduced segmentation precision.

By incorporating EPLD (Method 2), model performance is markedly improved, with mIoU reaching 66.16% and 58.47% on task POT IRRG → VAI IRRG and POT RGB → VAI IRRG, respectively. EPLD mitigates the negative influence of noisy pseudolabels through a dynamic entropy-based weighting mechanism, emphasizing high-confidence pseudolabels while suppressing noise during training. This enhances the model's capability to accurately capture fine-grained details in the target domain, resulting in more precise segmentation across several categories, including building and low vegetation.

Furthermore, compared with Method 1, incorporating EMIC (Method 3) also yields additional performance gains, achieving an mIoU of 65.31% and 55.43% on task POT IRRG → VAI IRRG and POT RGB → VAI IRRG, respectively. EMIC leverages masked high-entropy regions and contextual information for consistent prediction, thereby substantially enhancing the model's understanding of target-domain images. Through this masked-context reasoning, the model can infer predictions in uncertain regions, improving accuracy over using EPLD alone.

Overall, the proposed EntSeg method, combining EPLD and EMIC, achieves the optimal segmentation performance with an mIoU of 72.07% and 63.80% on task POT IRRG → VAI IRRG and POT RGB → VAI IRRG, respectively. These results demonstrate that jointly using EPLD and EMIC not only minimizes the adverse impact of noisy pseudolabels but also exploits contextual learning to accurately infer target-region predictions, enhancing both adaptability and precision in the target domain.

In summary, the ablation results confirm that EPLD and EMIC play a critical role in improving model performance. EPLD significantly reduces the negative effects of noisy pseudolabels via entropy-weighted learning, while EMIC further improves prediction accuracy in high-entropy, hard-to-predict regions through masked-context consistency. EntSeg effectively combines these advantages to achieve state-of-the-art cross-domain RS segmentation results.

2) Ablation on EMIC Mask Ratio and Patch Size: To investigate the impact of different mask ratios and patch sizes on model performance, multiple experimental configurations were evaluated in a comprehensive ablation study. As shown in Fig. 5, both mask ratio and patch size significantly affect segmentation results, and selecting appropriate values can effectively improve model accuracy.

First, the effect of varying the mask ratio was examined. The results indicate that a mask ratio of 0.5 yields the best performance, providing the most accurate predictions. Both higher ratios (e.g., 0.75) and lower ratios (e.g., 0.25) lead to inferior performance. A high mask ratio causes the model to rely excessively on contextual information, potentially losing

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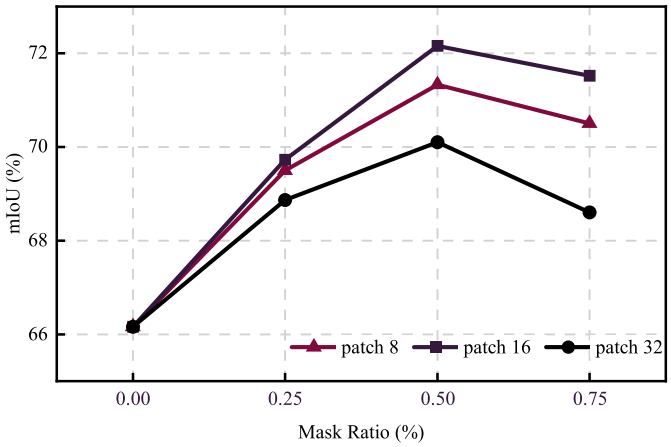


Fig. 5. Ablation results for different mask ratios and patch sizes. Mask ratios of 0%, 25%, 50%, and 75% were evaluated, with patch sizes of 8, 16, and 32. The figure shows experimental results on the Potsdam IRRG → Vaihingen IRRG task for these different configurations.

TABLE III
STUDY ON DIFFERENT WEIGHTING FORMULATIONS. THE ABERRATION PI→V STANDS FOR POT IRRG → VAI IRRG, WHILE PR→V STANDS FOR POT RGB → VAI IRRG

Method	Pi→V	Pr→V
No weighting	63.83	53.27
Exponential decay	71.36	63.44
Linear scale (ours)	**72.02**	**63.80**

local detail, whereas a low mask ratio prevents the model from fully leveraging context, limiting its global understanding.

The effect of different patch sizes was also analyzed. The experiments show that a patch size of 16 achieves optimal performance. At this size, the model can effectively capture local image features while maintaining robust global learning. A patch size of 16 balances fine-detail capture and computational efficiency, avoiding information loss or excessive overhead. On the contrary, smaller patches (e.g., 8) improve local detail capture but lose global context, while larger patches (e.g., 32) capture broader context but at the cost of detail resolution, reducing segmentation precision.

In conclusion, the optimal configuration is a mask ratio of 0.5 and a patch size of 16. This setting enables the model to exploit both local features and global context effectively, achieving the highest segmentation accuracy.

3) Study on Different Weighting Formulations: As shown in (8), the entropy weight $w_T^{i,h,w}$ in this work is obtained by applying a linear scaling to the entropy value $e_T^{i,h,w}$. A natural question is whether alternative mapping strategies, such as exponential decay, would influence the adaptation performance. To investigate this issue, we introduce an alternative weighting formulation based on exponential decay

$$w_T^{i,h,w} = \exp\left(-\lambda \frac{e_T^{i,h,w}}{\tau}\right) \tag{16}$$

where $\lambda = 1.0$ is a scaling factor and $\tau = \log(C)$ denotes the temperature parameter used to normalize the entropy values. This formulation assigns larger weights to low-entropy predictions while suppressing the influence of high-entropy (uncertain) pixels. Table III reports the comparison results using different weighting formulations. As observed, the performance differences between the linear scaling strategy and the exponential decay formulation are relatively small. This suggests that the primary performance gain mainly originates from the entropy-guided pseudolabel weighting mechanism itself, while the specific mapping function used to transform entropy values into weights has only a limited impact on the final performance. This observation indicates that the effectiveness of EPLD mainly comes from the entropy-guided reliability modeling rather than the specific functional form used for weight computation.

TABLE IV
STUDY ON DIFFERENT MASKING POLICIES. RESULTS ARE REPORTED ON POT IRRG → VAI IRRG

	Guidance	Mask level	mIoU (%)
1	Entropy	pixel	62.13
2	Random	patch	66.78
3	Entropy	patch	72.02

4) Study on Different Masking Policies: In the proposed method, the top-k high-entropy regions are masked at the patch level to perform masked-image consistency prediction. To further investigate the impact of different masking strategies, we conduct additional experiments by varying both the guidance signal and the masking granularity. In particular, we consider the following variants.

1) Entropy-guided pixel-level masking, where individual pixels with high entropy are masked independently.
2) Random patch-level masking, where patches are randomly selected without entropy guidance.
3) Entropy-guided patch-level masking (default setting).

The results are reported in Table IV. When entropy-guided masking is applied at the pixel level, the performance drops significantly. This degradation is likely caused by the disruption of spatial continuity, since independently masking pixels destroys the local spatial structure that is crucial for semantic segmentation. When random patch masking is adopted, the performance improves slightly compared to pixel-level masking but remains substantially lower than the entropy-guided masking strategy. This indicates that randomly masking regions does not effectively focus the learning process on unreliable predictions. On the contrary, the proposed entropy-guided patch masking achieves the best performance. By masking spatially coherent regions with high prediction uncertainty, the model is encouraged to learn more robust representations and improve prediction consistency under challenging regions.

5) Impact of the Number of Training Samples: To analyze the impact of the number of training samples, as shown in Table V, we conducted additional experiments by reducing the number of labeled source-domain samples. The results show that although the overall segmentation accuracy decreases

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TABLE V
IMPACT OF THE NUMBER OF TRAINING SAMPLES. RESULTS ARE REPORTED ON POT IRRG → VAI IRRG

Method	Sample ratio	mIoU (%)
DAFormer [23]	100%	63.25
SiamSeg [9]	100%	65.55
EntSeg (ours)	50%	66.43
EntSeg (ours)	75%	69.16
EntSeg (ours)	100%	72.07

when fewer labeled samples are used, the proposed EntSeg framework still consistently outperforms the DAFormer and SiamSeg. This observation indicates that the proposed entropy-guided learning strategy effectively improves the utilization of unlabeled target-domain data, making the framework more robust to limited labeled training samples.

6) Performance-Speed Tradeoff: As shown in Table VI, we compare DAFormer, SiamSeg, and EntSeg in terms of the number of parameters, GPU memory usage, and training/inference speed. The results indicate that EntSeg introduces a moderate increase in memory consumption and training time compared with the baseline methods. However, this additional computational overhead leads to a substantial improvement in segmentation accuracy. The performance gains demonstrate that EntSeg is an effective approach, significantly enhancing the model's generalization ability while incurring only a relatively small increase in computational cost.

V. DISCUSSION

A. Difference Between EntSeg and General UDA Methods

Compared with general UDA methods, which typically focus on feature distribution alignment (e.g., adversarial learning) or pseudolabeling (e.g., ST), the proposed EntSeg introduces an entropy-guided learning framework that explicitly addresses the reliability of pseudolabels and the learning of uncertain regions. First, conventional UDA methods usually treat pseudolabels equally or rely on simple confidence thresholding. On the contrary, EntSeg introduces the EPLD module, which dynamically evaluates prediction entropy and assigns pixel-wise weights to suppress the influence of noisy pseudolabels. Second, existing UDA approaches often do not explicitly address insufficient learning in ambiguous regions. EntSeg proposes the EMIC module, which masks high-entropy regions and encourages the model to infer them from contextual information, thereby improving representation learning in difficult areas. Finally, unlike general UDA methods that focus on a single aspect (e.g., alignment or pseudolabeling), EntSeg integrates entropy information into both pseudolabel learning and contextual consistency learning, forming a unified framework tailored for cross-domain RS semantic segmentation. Therefore, EntSeg is not a direct extension of general UDA methods, but a distinct technical framework based on uncertainty-aware learning, specifically designed to address noisy pseudolabels and insufficient learning under domain shift.

B. Discussion on EPLD

Although ADVENT [33] utilizes entropy to enhance domain adaptation performance, it is important to note that ADVENT is fundamentally an adversarial learning method. In ADVENT, entropy is mainly employed as an optimization objective to regularize the prediction distribution of the target domain through adversarial training. In particular, entropy minimization is used to encourage confident predictions and facilitate distribution alignment between domains. On the contrary, the proposed EPLD module adopts a fundamentally different strategy. During training, we dynamically estimate the entropy of predictions at each pixel location and explicitly use the entropy values to guide the learning weights assigned to pseudolabels. In other words, entropy is not used as a regularization objective but rather as a reliability indicator for pseudolabels, enabling the model to suppress the influence of noisy predictions in high-entropy regions while emphasizing more reliable low-entropy predictions. This design allows the model to mitigate the negative impact of noisy pseudolabels caused by domain shift while maintaining effective learning from reliable target-domain predictions.

C. Discussion on EMIC

It is true that MIC [53] introduces a masked image consistency strategy. However, the masking strategy adopted in MIC is purely random and does not consider the reliability of target-domain predictions. In UDA for semantic segmentation, pseudolabels generated on the target domain inevitably contain noise due to the domain discrepancy between the source and target domains. As a result, the predictions in many target regions exhibit high uncertainty and unreliable supervision signals. Random masking, as used in MIC, does not explicitly address this issue since it treats all pixels equally regardless of their prediction confidence. On the contrary, the proposed entropy-guided masking strategy explicitly leverages prediction entropy to identify unreliable regions. In particular, high-entropy regions are more likely to correspond to uncertain predictions and insufficiently learned target features. Therefore, instead of randomly masking image regions, our method intentionally masks high-entropy areas and encourages the model to infer their predictions from surrounding low-entropy contextual information. This design forces the network to better exploit contextual relationships within the image and improves representation learning for difficult target regions. Moreover, the proposed EMIC module is designed to work in conjunction with the EPLD module. While EPLD suppresses the negative impact of noisy pseudolabels through entropy-based reliability weighting, EMIC further improves target-domain learning by encouraging consistency between masked and full-image predictions. This complementary design alleviates the learning imbalance introduced by entropy-based weighting and promotes more effective adaptation to the target domain. Therefore, the proposed EMIC module is not a simple replacement of random masking with entropy-based masking. Instead, it introduces a targeted uncertainty-aware masking strategy tailored for UDA, which explicitly addresses the challenges of noisy

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TABLE VI
COMPUTATIONAL COST AND PERFORMANCE COMPARISON. WE REPORT PARAMETERS, TRAINING GPU USAGE (GB), AND TRAIN AND INFER SPEED (S/ITER). RESULTS ARE REPORTED ON POT IRRG → VAI IRRG

Method	Backbone	Params	Memery (GB)	Train (s/iter)	Infer(s/iter)	mIoU (%)
DAformer [23]	MiT-B5	89.9M	**8.31**	**0.55**	0.08	63.25
SiamSeg [9]	MiT-B5	89.9M	10.29	0.69	0.08	65.55
EntSeg (ours)	MiT-B5	89.9M	11.78	0.66	**0.08**	**72.07**

pseudolabels and insufficient learning in high-uncertainty regions.

D. EMA Smoothing Factor

In our implementation, the teacher network is updated using an EMA of the student network parameters, following the standard practice adopted in previous UDA semantic segmentation works such as [22] and [23]. In particular, we adopt the commonly used EMA smoothing factor $\alpha = 0.99$, which has been widely validated to provide stable pseudolabel updates in teacher–student frameworks. A larger α allows the teacher model to evolve more smoothly, thereby producing more stable pseudolabel predictions during training. Under large domain shifts, training stability is further supported by two key components in our framework. First, the proposed EPLD reduces the influence of unreliable predictions by assigning smaller weights to high-entropy regions. Second, the EMIC module encourages the model to learn more robust representations from uncertain regions. Together, these mechanisms effectively stabilize training even under significant domain discrepancies.

E. Compare With SiamSeg

Although both SiamSeg [9] and the proposed EntSeg address UDA for RS semantic segmentation, they are developed from different research perspectives. SiamSeg focuses on improving target-domain representation learning through contrastive learning. By constructing a Siamese framework, it enhances feature consistency across domains and alleviates domain discrepancies via contrastive objectives. On the contrary, EntSeg investigates the problem from the perspective of pseudolabel reliability and uncertainty-aware learning. In particular, EntSeg aims to mitigate the negative impact of noisy pseudolabels and insufficient learning in high-uncertainty regions. To this end, we introduce the EPLD module and the EMIC strategy, which together improve the robustness of pseudolabel learning and contextual reasoning in the target domain.

F. Limitations

Although the proposed EntSeg framework effectively mitigates performance degradation caused by the absence of target-domain labels, several limitations remain. First, similar to most UDA methods, EntSeg requires joint training on labeled source-domain data and unlabeled target-domain data. This implies that a separate model needs to be trained for each cross-domain task, which increases training overhead. Moreover, the reliance on source-domain data may lead to longer training time, especially when the source dataset is large or computational resources are limited. Second, the proposed entropy-guided learning strategy relies on prediction entropy as an indicator of pseudolabel reliability. However, entropy may not always perfectly reflect prediction uncertainty, particularly in complex scenarios where the model produces overconfident but incorrect predictions. In such cases, the effectiveness of entropy-based weighting may be limited. Third, the entropy-guided masking strategy in EMIC focuses on high-entropy regions to enhance contextual learning. While this design improves robustness, excessive masking or inaccurate identification of uncertain regions may reduce the available information for training, potentially affecting performance in extreme cases. Nevertheless, it is important to emphasize that the proposed method requires no target-domain labels and demonstrates strong adaptability in fully unsupervised scenarios, making it suitable for practical RS applications.

G. Future Works

Future research will explore source-free domain adaptation (SFDA), particularly under scenarios where source-domain data are unavailable during adaptation. SFDA eliminates the need to reuse source-domain data for each task, making it more aligned with real-world constraints in which source data may be inaccessible or restricted. Furthermore, we plan to investigate strategies to enhance the generalization capability of the model, enabling efficient adaptation across diverse cross-domain tasks while further reducing dependence on source data. These improvements aim to lower training costs and increase the practical flexibility of the proposed framework.

VI. CONCLUSION

This work introduced EntSeg, a framework designed to investigate and address the effects of noisy pseudolabels and low-confidence regions on cross-domain semantic segmentation. To alleviate the negative impact of noisy labels during adaptation, we proposed EPLD, which reweights each pixel based on its entropy, thereby assigning lower importance to noisy or high-entropy pixels. In addition, because high-entropy regions typically exhibit unreliable predictions and insufficient learning, we further proposed EMIC, which performs EMIC learning. EMIC encourages the model to infer masked high-entropy regions using contextual information rather than relying only on their raw appearance, thereby improving prediction reliability. Extensive experiments demonstrate that EntSeg, integrating both EPLD and EMIC, achieves substantially higher mIoU and accuracy compared with existing methods.

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